

Presynaptic nerve terminals release neurotransmitters via synaptic vesicle exocytosis. This action is mediated by the membrane-bound target-SNARE (t-SNARE) proteins syntaxin and SNAP-25, and the vesicle-associated SNARE (v-SNARE) protein synaptobrevin. Membrane fusion is initiated when the t-SNAREs and v-SNAREs form an α -helix bundle known as the core-*trans*-complex, achieved by zipping SNARE proteins together to form a more stable *cis*-complex (Figure 1).

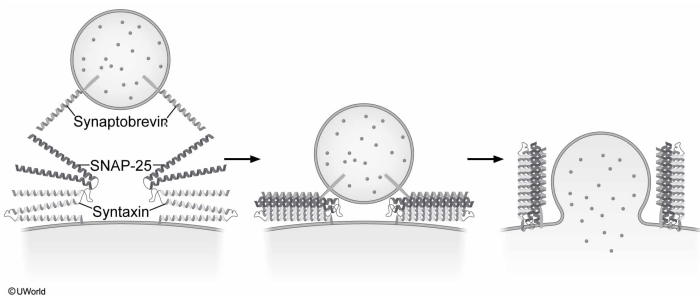


Figure 1 Formation of the core-*trans*-complex and subsequent membrane fusion

The *cis*-complex contains a region called the zero ionic layer, which is the main site of interaction between complexed proteins. Within the zero ionic layer, arginine on synaptobrevin coordinates with carbonyl groups present on the other residues shown in Table 1. The zero ionic layer is buried within amphipathic domains called leucine zippers, which shield the zero ionic layer from solvent molecules.

Table 1 Key Amino Acids of the *cis*-Complex Zero Ionic Layer

SNARE protein	Zero ionic layer residue
Synaptobrevin	R56
Syntaxin	Q226
SNAP-25 (Sn1)	Q53
SNAP-25 (Sn2)	Q174

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Once membrane fusion occurs, the *cis*-complex must disassemble. An early step in *cis*-complex disassembly is destabilization by the ATPase *N*-

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ethylmaleimide-sensitive factor (NSF), which disrupts leucine zipper regions, leading to vesicle reuptake. Movement of NSF to the cell membrane is mediated by the cytoplasmic protein α -SNAP, which recruits NSF after binding the N-terminal domain of syntaxin.

To determine the effect of synaptobrevin on the binding of α -SNAP to syntaxin, increasing amounts of recombinant His-tagged α -SNAP were added to a constant amount of syntaxin affixed to glutathione-agarose beads in the presence or absence of synaptobrevin. After incubation, bound α -SNAP was recovered and quantitatively analyzed by immunoblotting. Results are shown in Figure 2.

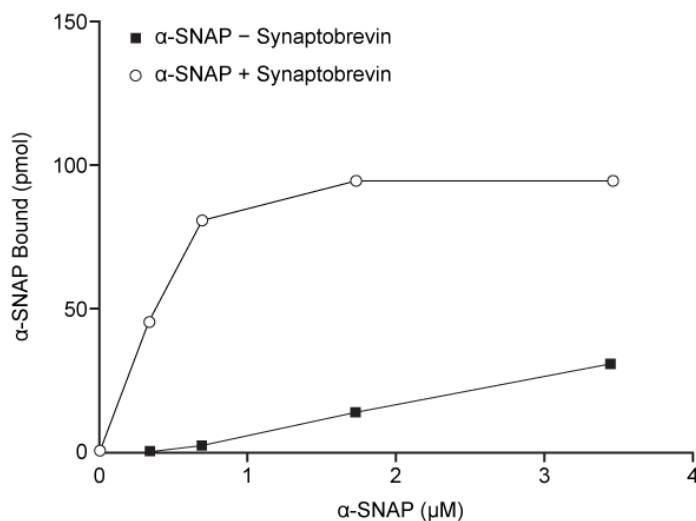


Figure 2 α -SNAP binding to syntaxin in the absence (square) or presence (circle) of synaptobrevin

Which of the following amino acid substitutions is most likely to interrupt the interactions of the zero ionic layer?

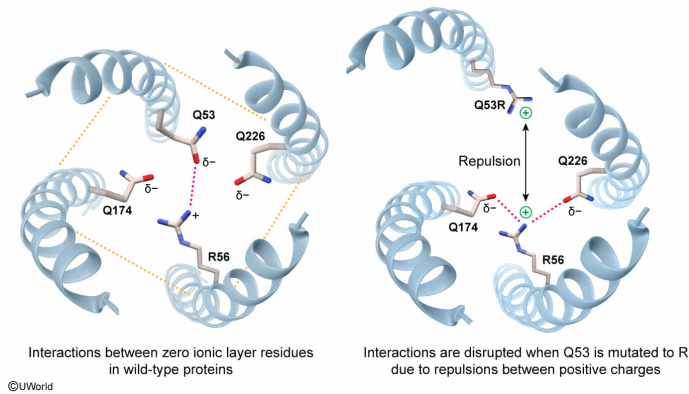
- ☐ A. R56K on synaptobrevin [17%]
- ☒ B. Q53R on SNAP-25 (Sn1) [61%]
- ☐ C. E224D on syntaxin [13%]
- ☐ D. Q174N on SNAP-25 (Sn2) [7%]

Correct

62% answered correctly

Explanation:

Explanation:



Amino acid substitutions can alter the behavior of the associated protein. Most commonly, these substitutions are denoted by writing the one-letter code of the original amino acid (ie, the one that is found in the wild-type protein) followed by the amino acid position, and then the one-letter code of the new amino acid. For instance, R56K indicates that the protein originally contained an arginine (R) residue at position 56 but this residue has been changed to lysine (K). Amino acid substitutions that alter charge are likely to destabilize protein interactions.

In this example, **Q53R** denotes the **replacement** of **glutamine** with **arginine** at **position 53** on SNAP-25 (Sn1). As the passage suggests, the zero ionic layer is stabilized by attractive forces between the carbonyl groups (partial negative charge) on **glutamine residues** and the positively charged **arginine** on synaptobrevin. Substitution of glutamine with arginine results in a **repulsive positive-positive ion interaction** between the two arginine residues, **destabilizing the interaction** between the proteins in the complex.

(Choice A) R56K denotes the substitution of arginine with **lysine** on synaptobrevin. Because both amino acids are positively charged, this substitution would be expected to have a relatively small effect on protein-protein interactions.

(Choice C) E224D denotes the substitution of **glutamate** at position 224 on syntaxin with **aspartate**. According to Table 1, residue 226 rather than 224 is in the zero ionic layer. Because glutamate and aspartate are both negatively charged, this substitution is unlikely to significantly alter the properties of the protein.

(Choice D) Q174N denotes the substitution of glutamine with **asparagine** on SNAP-25 (Sn2). Asparagine is similar in structure to glutamine, and therefore most likely interacts similarly within the zero ionic layer.

Educational objective:

Amino acid substitutions are denoted by the original amino acid, the amino acid position, and the substituting amino acid. Amino acid substitutions that alter charge are likely to destabilize protein interactions, particularly if they result in repulsive forces between similarly charged amino acids.

Time spent:0 sec

Subject:
Biochemistry

QID:400070

System:
Amino Acids and Proteins

Presynaptic nerve terminals release neurotransmitters via synaptic vesicle exocytosis. This action is mediated by the membrane-bound target-SNARE (t-SNARE) proteins syntaxin and SNAP-25, and the vesicle-associated SNARE (v-SNARE) protein synaptobrevin. Membrane fusion is initiated when the t-SNAREs and v-SNAREs form an α -helix bundle known as the core-*trans*-complex, achieved by zipping SNARE proteins together to form a more stable *cis*-complex (Figure 1).

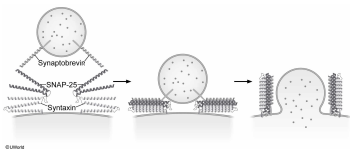


Figure 1 Formation of the core-*trans*-complex and subsequent membrane fusion

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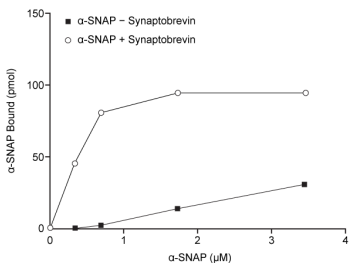


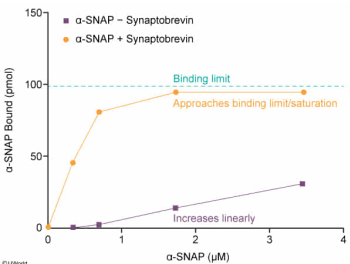
Figure 2 α -SNAP binding to syntaxin in the absence (square) or presence (circle) of synaptobrevin

Which of the following statements regarding the binding of α -SNAP to syntaxin is best supported by the results shown in Figure 2?

- ☐ A. Binding does not occur in the presence of synaptobrevin.
- ☐ B. Synaptobrevin induces a conformation change in syntaxin that results in lower binding affinity. [9%]
- ☒ C. In the absence of synaptobrevin, binding does not saturate under the conditions of the experiment. [79%]
- ☐ D. Interaction of synaptobrevin and syntaxin raises the K_d for the dissociation of syntaxin from α -SNAP. [10%]

Correct
79% answered correctly

Explanation:



Substrate binding curves relate the number of bound substrate molecules (percent saturation) to substrate concentration. **Initially**, substrate-protein binding is **approximately linear**. When proteins reach **saturation**, all **binding sites are occupied**; this condition can be observed as the binding curve **leveling off** at the binding limit, which is dependent on the amount of protein present and the number of binding sites. At this binding limit, addition of more substrate does not result in increased binding.

Figure 2 shows that when synaptobrevin is present, increasing α -SNAP beyond 2 μM does not result in increased binding, suggesting that all binding sites are saturated. Conversely, when synaptobrevin is absent, α -SNAP binding to syntaxin is reduced, and the amount of α -SNAP bound to syntaxin **increases linearly** with increasing α -SNAP concentration over the range tested, suggesting that **binding has not approached saturation**.

(Choice A) The data show that α -SNAP binds to syntaxin to different extents in both the presence and absence of synaptobrevin.

(Choice B) The amount of α -SNAP bound to syntaxin *increases* when synaptobrevin is present. This behavior suggests a *higher* binding affinity in the presence of synaptobrevin.

(Choice D) The dissociation constant K_d is an equilibrium constant for the dissociation of bound molecules into two or more unbound molecules. Increased K_d reflects *decreased* binding affinity. The data in Figure 2 suggest the opposite, that synaptobrevin interacting with syntaxin *increases* α -SNAP binding affinity and *decreases* the K_d for α -SNAP dissociation.

Educational objective:

On substrate binding curves, substrate-protein binding is approximately linear at low substrate levels. When proteins reach saturation, all binding sites are occupied and binding curves level off because the further addition of substrate does not increase binding.

Time spent: 0 sec
Subject:
Biochemistry

QID: 400071
System:
Amino Acids and Proteins

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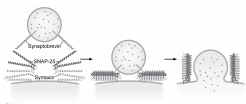


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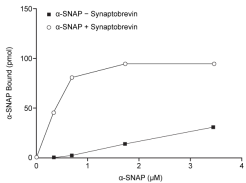


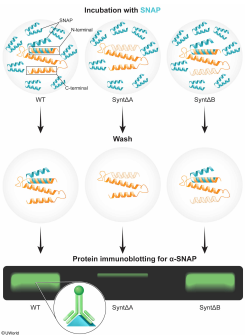
Figure 2 α -SNAP binding to syntaxin in the absence (square) or presence (circle) of synaptobrevin

Wild-type syntaxin (Synt-WT) and mutant syntaxins containing a deletion of the N-terminal sequence (Synt Δ A) or C-terminal sequence (Synt Δ B) were separated and incubated with α -SNAP. After washing away unbound molecules, which of the following is the most likely result of protein immunoblotting for α -SNAP?

- ☒ A.
- | GST-Synt1A-WT | GST-Synt1 Δ A | GST-Synt1 Δ B |
|---------------|----------------------|----------------------|
| | | |
- [69%]
- ☐ B.
- | GST-Synt1A-WT | GST-Synt1 Δ A | GST-Synt1 Δ B |
|---------------|----------------------|----------------------|
| | | |
- [3%]
- ☐ C.
- | GST-Synt1A-WT | GST-Synt1 Δ A | GST-Synt1 Δ B |
|---------------|----------------------|----------------------|
| | | |
- [23%]
- ☐ D.
- | GST-Synt1A-WT | GST-Synt1 Δ A | GST-Synt1 Δ B |
|---------------|----------------------|----------------------|
| | | |
- [3%]

Correct
69% answered correctly

Explanation:



Western blot (protein immunoblotting) is a protein separation and identification technique that uses labeled **antibodies** to **detect specific proteins** after **gel electrophoresis**. In this example, the antibody is designed to bind α -SNAP. Therefore, after unbound α -SNAP is washed away, protein samples are visualized on western blot in **proportion to α -SNAP binding**.

The passage indicates that α -SNAP binds syntaxin at its N-terminal domain. The mutant syntaxin Synt Δ A described in the question lacks the N-terminal domain and is unlikely to bind α -SNAP in large quantities, leading to little α -SNAP-bound syntaxin for this mutation. In contrast, deletion of the C-terminal domain (Synt Δ B) is not expected to affect α -SNAP binding compared to binding of the wild-type (WT) protein. Therefore, α -SNAP in the WT protein and Synt Δ B samples are visualized as large bands whereas **less α -SNAP** is visualized in the **Synt Δ A sample**.

(Choice B) α -SNAP poorly binds the mutant syntaxin Synt Δ A. The quantity (band intensity) of α -SNAP-bound Synt Δ A should be less than WT and Synt Δ B.

(Choice C) According to the passage, α -SNAP binds syntaxin's N-terminal domain. Modification of the C-terminal domain in Synt Δ B should not impact α -SNAP binding, so the Synt Δ B lane should look similar to the WT lane.

(Choice D) Synt-WT binds α -SNAP effectively, so the western blot should show a full band in the WT lane.

Educational objective:

Western blot (protein immunoblotting) uses antibodies to detect specific proteins. Greater quantities of the protein of interest manifest as larger bands on the resultant blot.

Time spent:0 sec
Subject:
Biochemistry

QID:400072
System:
Amino Acids and Proteins